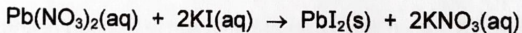


- 18 When 100 cm^3 of 1.0 mol dm^{-3} $\text{Pb}(\text{NO}_3)_2$ and 100 cm^3 of 1.0 mol dm^{-3} KI are mixed together, the temperature changes from 21.5°C to 26.6°C .

Assume that the specific heat capacity of the solution is $4.18\text{ J g}^{-1}^\circ\text{C}^{-1}$ and that the density of the solution is 1.00 g cm^{-3} .



What is the $\Delta H_{\text{reaction}}$ per mole of PbI_2 formed?

- A -85.3 kJ mol^{-1}
- B -42.6 kJ mol^{-1}
- C -21.3 kJ mol^{-1}
- D $+42.6\text{ kJ mol}^{-1}$